```
import java.util.Scanner;

public class SmartParkingSystem {

static Scanner sc = new Scanner(System.in);
static boolean[] slots = new boolean[6];

static void userInput() {
System.out.print("Enter your name: ");
String name = sc.nextLine();

System.out.print("Enter vehicle number: ");
String vehicle = sc.nextLine();

System.out.println("Welcome, " + name);
System.out.println("Vehicle Number: " + vehicle);
}

static void retrieveSlots() {
System.out.println("\n--- Parking Slots ---");
```

```
for (int i = 1; i <= 5; i++) {
if (!slots[i]) {
System.out.println("Slot " + i + " is Available");
} else {
System.out.println("Slot " + i + " is Occupied");
}
}
}

static void validateSlot() {
System.out.print("Enter slot number to book
(1-5): ");
int slot = sc.nextInt();

if (slot < 1 || slot > 5) {
System.out.println("Invalid slot number!");
} else if (slots[slot]) {
System.out.println("This slot is already
occupied!");
```

```
} else {
slots[slot] = true;
System.out.println("Slot " + slot + " booked
successfully!");
}
}

static void displayStatus() {
int available = 0;
int occupied = 0;

for (int i = 1; i <= 5; i++) {
if (slots[i]) {
occupied++;
} else {
available++;
}
}

System.out.println("\n--- Parking Status ---");
```

```
System.out.println("Total Slots: 5");
System.out.println("Available Slots: " + available);
System.out.println("Occupied Slots: " + occupied);
}

static void updateRecord() {
System.out.print("Enter slot number to release
(1-5): ");
int slot = sc.nextInt();

if (slot < 1 || slot > 5) {
System.out.println("Invalid slot number!");
} else if (!slots[slot]) {
System.out.println("This slot is already empty!");
} else {
slots[slot] = false;
System.out.println(
"Parking record updated. Slot released!");
}
}
```

```
public static void main(String[] args) {
int choice;

do {
System.out.println("\n===== SMART PARKING
SYSTEM =====");
System.out.println("1. User Input");
System.out.println("2. View Parking Slots");
System.out.println("3. Book Parking Slot");
System.out.println("4. Display Parking Status");
System.out.println("5. Release Parking Slot");
System.out.println("6. Exit");
System.out.print("Enter your choice: ");

choice = sc.nextInt();
sc.nextLine();

switch (choice) {
case 1:
```

```
userInput();
break;

case 2:
retrieveSlots();
break;

case 3:
validateSlot();
break;

case 4:
displayStatus();
break;

case 5:
updateRecord();
break;

case 6:
```

```
System.out.println(
"Thank you for using Smart Parking!");
break;

default:
System.out.println(
"Invalid choice. Try again!");
}

} while (choice != 6);

sc.close();
}
}
```